

Justin Roberts

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EDUCATION

Cornell University

Ithaca, NY

Bachelor of Arts in Mathematics and Computer Science

Expected: May 2028

Relevant Coursework: Data Structures & Algorithms, Computer Systems, Linear Algebra, Probability

EXPERIENCE

Software Engineer Intern

June 2026 – Present

Wayfair

Boston, MA

- Built an end-to-end image upload and retrieval pipeline in Java/Spring Boot with GraphQL, GCS signed URLs, and DAM, supporting 10K+ uploads/day while maintaining sub-500 ms p95 latency
- Optimized PostgreSQL ingestion procedures processing 5M+ third-party reviews annually, reducing redundant UPDATE operations by 80% and downstream Airflow events by 60%
- Refactored a core Java review-summary parser to support an enriched format, increasing unit test coverage from 70% to 90%+ and reducing staging parsing errors by 40%

Software Engineer Intern

Sept 2025 – May 2026

PlayToon

Remote

- Redesigned and optimized Prisma ORM database schema in JavaScript that reduced data redundancy by 30%
- Designed an AWS S3 upload pipeline proposal comparing presigned URL and server-mediated flows across dev/staging/prod, identifying a path to reduce failed uploads by 40% and simplify IAM role management by 50%
- Overhauled and expanded developer documentation with setup guides and architecture diagrams, cutting onboarding time by 30% and increasing cross-team code consistency by 20%

Independent Researcher

May 2025 – Oct 2025

Stony Brook University

Stony Brook, NY

- Developed sublinear graph algorithms using global queries to detect isolated vertices, achieving logarithmic query complexity while reducing query complexity by 75% over prior approaches
- Validated testing performance using Python on 1,000+ synthetic graph datasets, achieving 25% faster runtime than legacy implementations

PROJECTS

Buzzword Decoder | Demo | *JavaScript, Express, Chrome Ext. API, Claude API*

July 2026

- Built a Chrome extension (Manifest V3) and a Node/Express proxy on Render integrating Claude Haiku 4.5 to translate corporate jargon in real time, achieving sub-2s end-to-end latency per page
- Engineered a whitespace-normalization algorithm that maps collapsed webpage text back to original DOM offsets, increasing phrase-match accuracy by ~35% across scanned pages
- Secured the Anthropic API key behind a Render-hosted, token-authenticated proxy with constant-time comparison, eliminating client-side exposure at under 100ms p95 overhead
- Isolated all extension styling in a Shadow DOM with live per-tab toolbar state indicators, delivering zero CSS conflicts across 100+ manually tested webpages

Trading Bot | *Python, yfinance API, Alpha Vantage API, Git*

Aug 2025 – Jan 2026

- Developed a Python trading signal engine implementing SMA, Bollinger Bands, and RSI indicators to generate automated buy/sell recommendations
- Built a market data pipeline integrating yfinance, Alpha Vantage, and ccxt APIs to support automated signal generation across equities and cryptocurrencies
- Generated 21.50% gain over the course of 2025 on BTC, 14.39% on ETH, and 16.29% on BNB

TECHNICAL SKILLS

Languages: Java, Python, C, JavaScript, SQL, GraphQL, Assembly, Julia, HTML/CSS, PHP

Frameworks: Spring Boot, React, Node.js, Express, FastAPI, Django, PyTorch

Developer Tools: AWS, GCS, Docker, Kubernetes, Git, GitHub, DataDog, Buildkite, Chrome Extension APIs (Manifest V3), Anthropic API

Libraries: pandas, NumPy, Matplotlib, ccxt, yfinance, Alpha Vantage